Result

The information of the CV:

```
Here's a breakdown of the CVs and the candidates they represent:

**1. Rahul Sharma (AI Professional)**

*   **Profession:** AI Professional
*   **Skills:** Machine Learning & AI (Advanced AI Systems, ML, NLP), Frameworks & Tools (TensorFlow, PyTorch, Py
*   **Education:** PhD in Artificial Intelligence, University of Tokyo (2012)
*   **Experience:**
    *   Senior Engineer, EY (Current)
    *   Consultant, StartupXYZ (2019-2021)
    *   Senior Analyst, DataForge (2016-Present)
    *   Lead Scientist, Urban Flow (2010-2017)
*   **Locations:** London, Hong Kong, Singapore (Hometown)

**2. Minh Pham (Design Professional)**

*   **Profession:** Design Professional
*   **Skills:** UI/UX Design, Prototyping, Graphic Design
*   **Education:** BSc in Design, The University of Hong Kong (2011)
*   **Experience:**
    *   Manager, BCG (2022-Present)
    *   Analyst, Tencent (2013-2017)
*   **Locations:** Beijing, China | Hong Kong

**3. Rahul Sharma (Legal Professional)**

*   **Profession:** Legal Professional
*   **Skills:** Compliance, Litigation, Contract Review, Web3, Machine Learning, Quantum Computing
*   **Education:** PhD in Legal Studies, Tsinghua University (2021)
*   **Experience:**
    *   Senior Engineer, Microsoft (2021-2027)
    *   Consultant, StartupXYZ (2020-2023)
*   **Locations:** Singapore (Hometown), Singapore / Philippines

**4. Wei Zhang (Consulting Professional)**
```

```
**Candidates Identified:**

1.  Rahul Sharma (AI Professional)
2.  Minh Pham (Design Professional)
3.  Rahul Sharma (Legal Professional)
4.  Wei Zhang (Consulting Professional)
5.  John Smith (Marketing Professional)
```

Calling MCP:

```
Iteration 1: Invoking LLM...
Tool calls detected: 3
  -> Calling tool: search_linkedin_people with args: {'q': 'Rahul Sharma', 'location': 'London'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":12,"name":"Rahul Sharma","headline":"Logistics Professional","indu...
  -> Calling tool: search_linkedin_people with args: {'industry': 'AI', 'location': 'London', 'q': 'Rahul Sharma'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":5941,"name":"Rahul Sharma","headline":"AI Professional","industry"...
  -> Calling tool: search_facebook_users with args: {'q': 'Rahul Sharma'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":4,"display_name":"Rahul Sharma","city":"Berlin","country":"Germany...
Iteration 2: Invoking LLM...
Tool calls detected: 2
  -> Calling tool: get_linkedin_profile with args: {'person_id': 5941}
  <- Tool output: [{'type': 'text', 'text': '{"id":5941,"name":"Rahul Sharma","headline":"AI Professional","city":"Lon...
  -> Calling tool: get_facebook_profile with args: {'user_id': 12}
  <- Tool output: [{'type': 'text', 'text': '{"id":12,"display_name":"Rahul Sharma","original_name":"Rahul Sharma","ci...
Iteration 3: Invoking LLM...
Tool calls detected: 2
  -> Calling tool: search_linkedin_people with args: {'location': 'Hong Kong', 'q': 'Minh Pham'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":180,"name":"Minh Pham","headline":"Consulting Professional","indus...
  -> Calling tool: search_facebook_users with args: {'q': 'Minh Pham'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":62,"display_name":"Minh Pham","city":"Austin","country":"USA","mat...
Iteration 4: Invoking LLM...
Tool calls detected: 2
  -> Calling tool: get_linkedin_profile with args: {'person_id': 2758}
  <- Tool output: [{'type': 'text', 'text': '{"id":2758,"name":"Minh Pham","headline":"Design Professional","city":"Ca...
  -> Calling tool: get_facebook_profile with args: {'user_id': 180}
  <- Tool output: [{'type': 'text', 'text': '{"id":180,"display_name":"Minh Pham","original_name":"Minh Pham","city":"...
Iteration 5: Invoking LLM...
Tool calls detected: 2
  -> Calling tool: search_linkedin_people with args: {'location': 'Singapore', 'q': 'Rahul Sharma'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":547,"name":"Rahul Sharma","headline":"Logistics Professional","ind...
  -> Calling tool: search_facebook_users with args: {'q': 'Rahul Sharma'}
  <- Tool output: [{'type': 'text', 'text': '[{"id":4,"display_name":"Rahul Sharma","city":"Berlin","country":"Germany...
Iteration 6: Invoking LLM...
Tool calls detected: 2
  -> Calling tool: get_linkedin_profile with args: {'person_id': 2919}
  <- Tool output: [{'type': 'text', 'text': '{"id":2919,"name":"Rahul Sharma","headline":"Legal Professional","city":"...
```

Verification Reports and Scores:

```
**1. Rahul Sharma (AI Professional)**

*   **CV Claims:**
    *   Profession: AI Professional
    *   Skills: Machine Learning & AI (Advanced AI Systems, ML, NLP), Frameworks & Tools (TensorFlow, PyTorch, Python)
    *   Education: PhD in Artificial Intelligence, University of Tokyo (2012)
    *   Experience: Senior Engineer, EY (Current), Consultant, StartupXYZ (2019-2021), Senior Analyst, DataForge (2016-Pr
t), Lead Scientist, Urban Flow (2010-2017)
    *   Locations: London, Hong Kong, Singapore (Hometown)
*   **LinkedIn Profile (ID: 5941):**
    *   Headline: AI Professional
    *   Industry: AI
    *   Skills: ML, TensorFlow, NLP, Python
    *   Experience: Engineer, GreenLeaf Co (2014-Present), Manager, PwC (2010-Present)
    *   Education: PhD, KAIST (2003-2006)
    *   Location: London
*   **Facebook Profile (ID: 12):**
    *   City: London
    *   Education: Doctoral Degree
    *   Current Job: Manager, PwC
*   **Discrepancies:**
    *   Education: CV claims PhD from University of Tokyo (2012), LinkedIn shows PhD from KAIST (2003-2006).
    *   Experience: CV lists several roles, LinkedIn shows Engineer at GreenLeaf Co and Manager at PwC.
    *   The dates of employment also do not match.
*   **Verification Score:** 0.3 (Significant discrepancies in education and experience)

**2. Minh Pham (Design Professional)**

*   **CV Claims:**
    *   Profession: Design Professional
    *   Skills: UI/UX Design, Prototyping, Graphic Design
    *   Education: BSc in Design, The University of Hong Kong (2011)
```

Test

1.When test the code,please change line 31 to your api.

```python
28
29      llm = ChatGoogleGenerativeAI(
30          model="gemini-2.0-flash",
31          api_key=os.getenv("v_api"),
32          vertexai=True,
33          temperature=0.2,
34      )
```

2.Put all CVs into cv directory,and the agent will indentify them.

3.If more CV need to be verificated,please change the turns in line 115.In avreage,each CV may need 4 turns.

```python
114
115     for i in range(20):
116         print(f"Iteration {i+1}: Invoking LLM...")
117         ai_msg = llm_with_tools.invoke(messages)
118         messages.append(ai_msg)
119
120         # 1. if there are tool calls, execute the tools
121         if ai_msg.tool_calls:
122             print(f"Tool calls detected: {len(ai_msg.tool_calls)}")
123             for tool_call in ai_msg.tool_calls:
124                 tool_name = tool_call["name"]
125                 tool_args = tool_call["args"]
126                 tool_call_id = tool_call["id"]
```